

Diversidad alfa por transecto de aridez

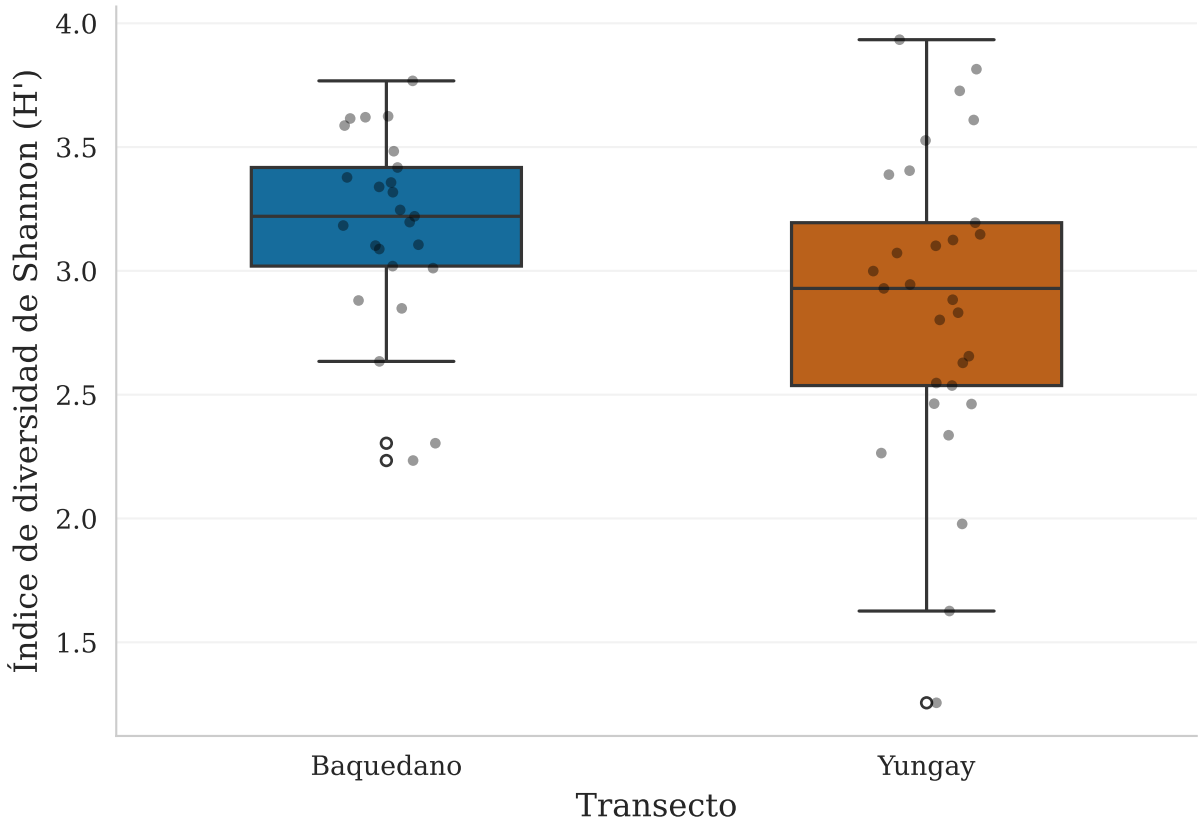


Figure 1. Alpha diversity of soil microbial communities across an aridity gradient in the Atacama Desert. Shannon diversity index (H') is shown for samples from the Baquedano transect (more humid, $n = 25$) and the Yungay transect (more arid, $n = 29$). Boxes show median and interquartile range; points represent individual samples. Baquedano exhibits higher median diversity than Yungay, consistent with reduced water availability filtering out desiccation-sensitive taxa.

Diversity increases with soil moisture availability

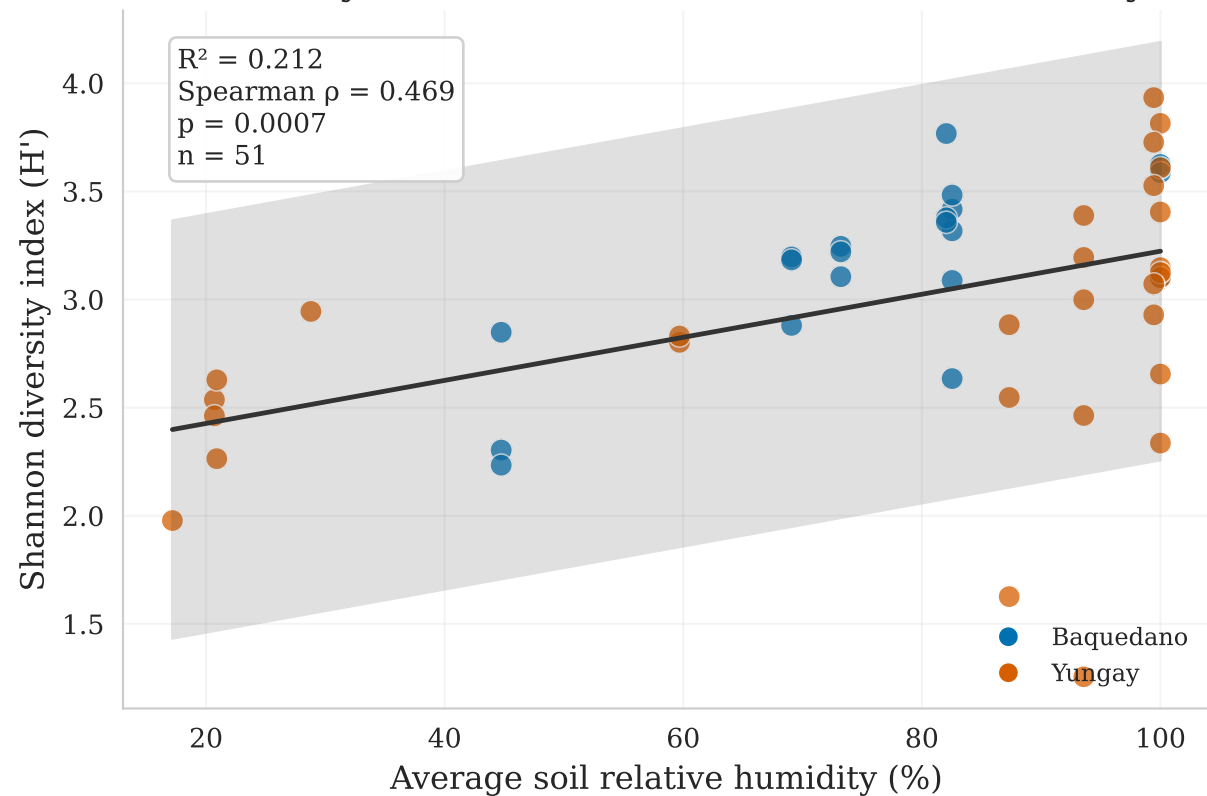


Figure 2. Relationship between soil moisture and microbial alpha diversity. Shannon diversity index plotted against average soil relative humidity for all samples with available humidity data ($n = 51$). Solid line shows the linear regression fit; shaded band indicates the 95% confidence interval. A significant positive correlation (Spearman $\rho = 0.47$, $p = 0.0007$) supports the hypothesis that increasing aridity reduces microbial diversity in Atacama Desert soils.